Gautham Ramkumar

Ashburn, VA | +1 (857) 565-5374 | ramkumar.g@northeastern.edu
linkedin.com/in/rgautham20 | github.com/gautham-ramkumar | gautham-ramkumar.github.io
Authorized to work in the US for 3 years with no sponsorship required

KEY ACHIEVEMENTS

- Placed 2nd at Revolute Hackathon, shipping a 3-model voice-to-action manipulation pipeline in one weekend

EDUCATION

Northeastern University, Boston, MA May 2026
Master of Science in Robotics (Concentration: ECE) *GPA: 3.636/4.0*

Vellore Institute of Technology, India May 2024
Bachelor of Technology in Electronics and Communications Engineering *GPA: 3.48/4.0*

TECHNICAL SKILLS

Languages	C++17, Python
Vision & 3D	3D Gaussian Splatting, Structure from Motion, Bundle Adjustment, COLMAP, OpenCV
Perception	Camera-LiDAR-IMU Calibration, Point Cloud Filtering (KD-tree), Stereo Vision (ZED Mini)
Deep Learning	PyTorch, CNNs / U-Net, Vision-Language Models, TensorRT, Edge Deployment
Software Practices	ROS2, ROS1 (Kalibr), Docker, Linux, Git, PyTest, GitHub Actions CI/CD, MuJoCo

PROFESSIONAL EXPERIENCE

Chicago Education Advocacy Corporation Aug 2026 – Present
Robotics AI Engineer, Chicago, IL (Remote)

- Grew automated test coverage from 44 to 145 tests across a two-job CI pipeline
- Reproduced a 70% SmolVLA task-success baseline in MuJoCo on an 8 GB GPU with pinned dependencies
- Built a dataset verifier that catches 3 silent defects (zeroed chunks, wrong state convention, no unnormalization)
- Built Backstop, a VLA failure monitor reaching 85.4% detection at 2.9% false-stop on held-out episodes

SmartInternz May 2023 – Jul 2023
Internet of Things Intern, Chennai, India

- Maintained 1 cm distance accuracy in a real-time ultrasonic sensing pipeline for automated access control

PROJECTS

3D Gaussian Splatting from Scratch Jan 2026 – Mar 2026
Python, PyTorch, CUDA, COLMAP, Docker | Code

- Implemented the full Gaussian optimization pipeline from scratch (densification, cloning, splitting, pruning) on the official EWA CUDA rasterizer
- Established 7,500 iterations as the optimal stopping point for small scenes at 31.42 dB PSNR / 0.9531 SSIM, over 16 dB better than the 30k checkpoint

DarkSight — Low-Light Perception 2025 – Present
PyTorch, TensorRT, RAFT, YOLOv8, RealSense | Code

- Designed a DS3DConv encoder-decoder with a ConvGRU bottleneck for frame memory and RAFT optical-flow warping for temporal consistency
- Reached 17.25 dB PSNR at 0.99 ORB stability, running 21 FPS at 360p with 57 ms latency on an Intel RealSense D435

3D Reconstruction via Structure from Motion 2025
Python, OpenCV, GTSAM, Open3D | Code

- Built an incremental SfM pipeline from scratch: SIFT features, RANSAC outlier rejection, essential-matrix decomposition, and triangulation
- Optimized roughly 5,500 landmarks through GTSAM bundle adjustment, cutting cost 60-70% and registering all 24 views at 3-4 px reprojection RMSE

Camera-LiDAR-IMU Extrinsic and Temporal Calibration Oct 2025 – Dec 2025
C++17, Python, ROS2, GTSAM, PyTest | Code

- Rejected an unstable 150ms camera-LiDAR offset; 748 timestamp pairs gave -19.5ms within 1.5ms over 15 windows
- Tripled usable LiDAR point density per scan via a KD-tree dynamic-point filter, no segmentation network